

Hand Gesture Classification on Edge Devices

Team 14

王昱閔 王辰竹 羅詠璿

Abstract

We present a compact multimodal hand-gesture classifier for edge deployment. The model receives only a cropped RGB hand image and 21 MediaPipe landmarks, then predicts one of five command gestures or rejects the input as N/A. A depthwise-separable convolutional branch captures appearance, while a landmark multilayer perceptron captures hand geometry. Their features are fused and followed by confidence-based rejection calibrated for an asymmetric evaluation rule in which a false trigger is penalized twice as heavily as a correct command is rewarded. The final network contains 36,039^[VERIFY] trainable parameters. Its tensors occupy 0.142 MiB^[VERIFY], while the submitted PyTorch checkpoint occupies 0.168 MiB^[VERIFY]. The best reported validation accuracy is 96.75%^[PROG], and the stronger of two leaderboard submissions obtained a presentation subtotal of 81.4/90^[PRES].

1 Introduction

Touch-free gesture interfaces must recognize intended commands while avoiding activation on ordinary hand poses. This requirement is especially important on edge devices, where models must also remain compact and operate without cloud inference. The course challenge fixes the upstream MediaPipe preprocessor and permits the downstream classifier to receive only a cropped RGB hand image and 21 hand landmarks [1, 2].

The task uses five valid commands—fist, like, OK, one, and palm—and maps every other accepted one-hand pose to N/A. Its scoring rule gives one point for a correct target prediction but subtracts two points for a false trigger [1]. This asymmetry changes the design objective: maximizing closed-set accuracy alone is insufficient, because uncertain samples should often be rejected.

Our main contributions are:

- a dual-stream model combining visual appearance and landmark geometry within a small parameter budget;
- a three-stage training procedure consisting of image pretraining, joint multimodal optimization, and scoring-aware calibration; and
- a conservative inference policy combining neural N/A prediction, confidence and margin gates, and landmark validity checking.

2 Task and Evaluation

2.1 Inputs and label space

The deployed function follows the required `predict(cropped_img, landmarks)` interface ^[INFER]. The image is resized to 64×64 ^[INFER] RGB pixels and normalized channel-wise from $[0, 1]$ to $[-1, 1]$. The 21^[SPEC] crop-relative (x, y) landmarks are flattened into a 42^[INFER]-dimensional vector. The output mapping is shown in Table 1. Full-frame images are prohibited [1].

2.2 Scoring rule

The official evaluation allocates 30 points to model size, 20 to HaGRIDv2 performance, 40 to real-world robustness, and 30 to presentation [1]. For a model no larger than 10 MB, the size score is

$$S_{\text{size}} = (10 - \text{size}_{\text{MB}}) \times 3. \quad (1)$$

Both performance sets use the raw contribution

$$s_i = \begin{cases} +1, & \text{correct target command,} \\ -2, & \text{false trigger,} \\ 0, & \text{N/A output.} \end{cases} \quad (2)$$

The robustness set contains 50 N/A interference images and 50 target images [1]. Since one false trigger cancels two correct commands, we optimize checkpoint selection and calibration with this scoring function in addition to reporting accuracy.

3 Related Work

MediaPipe Hands combines a palm detector and a landmark estimator for real-time on-device hand tracking [2]. Its landmark representation provides geometric information that is less dependent on texture and illumination than RGB appearance alone. HaGRIDv2 is a large-scale gesture dataset containing static, dynamic, and no-gesture examples [3]. Our classifier uses its 512-pixel release while following the challenge-specific target and N/A mapping.

To reduce computation, the image encoder uses depthwise-separable convolutions, a factorization used

Table 1: Output label mapping used by the submitted inference code.

ID	Label	ID	Label
0	N/A	3	OK
1	fist	4	one
2	like	5	palm

by MobileNets to replace a standard spatial convolution with depthwise and pointwise operations [4]. The proposed network is task-specific and trained from scratch; it does not use a gesture-recognition pretrained model.

4 Proposed Method

4.1 Dual-stream architecture

Figure 1 summarizes the model. The RGB crop and landmark vector are encoded independently, producing two 64-dimensional descriptors. These descriptors are concatenated into a 128-dimensional multimodal feature, which is classified into six logits. The image branch contains 13,249^[VERIFY] trainable parameters, the landmark branch contains 14,144^[VERIFY], and the fusion head contains 8,646^[VERIFY]. This division gives both modalities comparable model capacity while keeping the complete network small.

4.2 Image branch

The image branch contains five depthwise-separable blocks. For an input with C channels, the 3×3 depthwise convolution uses C groups and filters each channel independently. A 1×1 pointwise convolution then mixes information across channels and changes the channel count. Each operation is followed by batch normalization and ReLU [5]. This factorization replaces a dense spatial convolution while preserving its local receptive field.

A central design decision keeps the model far smaller than typical submissions: we reuse only the depthwise-separable *building block* introduced by MobileNet [4], rather than an entire MobileNet backbone. A complete MobileNetV2 or MobileNetV3 image encoder holds on the order of millions of parameters—roughly 3.4 M and 2.5 M for MobileNetV2 and MobileNetV3-Small, respectively—and is usually deployed by fine-tuning ImageNet-pretrained weights [6, 7]. In contrast, our image encoder stacks only five such blocks and is trained from scratch on the gesture data (Section 6), so it contains just 13,249^[VERIFY] parameters—about two orders of magnitude fewer than a full backbone, as summarized in Table 2. Two factors make this reduced capacity sufficient: the hand crop is already tightly localized by the fixed MediaPipe stage, and only six coarse classes must be separated. This choice, rather than post-hoc compression, is the primary reason the complete network occupies under 0.2 MiB instead of the several megabytes typical of a fine-tuned backbone, which directly raises the size component of the official score.

Table 3 gives the complete tensor progression. The first three blocks use stride two and reduce the spatial resolution from 64×64 to 8×8 ^[ARCH]; Blocks 4 and 5 refine the features without further downsampling. Adaptive global average pooling averages each of the final 64 feature maps over its 8×8 spatial grid, yielding one scalar per channel and therefore a 64-dimensional^[ARCH] image descriptor.

The branch is also inexpensive to evaluate. Summing

Table 2: Parameter-count comparison between our from-scratch image encoder and standard MobileNet image backbones. MobileNet counts are the published ImageNet encoder sizes [6, 7]; ours are recomputed from the model definition^[VERIFY].

Image encoder	Parameters
MobileNetV2 (full backbone)	$\sim 3,400,000$
MobileNetV3-Small (full backbone)	$\sim 2,500,000$
Ours (5 DW-sep blocks, from scratch)	13,249
Ours (full dual-stream model)	36,039

Table 3: Detailed image-branch architecture. Every block follows DWConv–BN–ReLU–PWConv–BN–ReLU.

Layer	Input tensor	Output tensor	Stride
Block 1	$64 \times 64 \times 3$	$32 \times 32 \times 16$	2
Block 2	$32 \times 32 \times 16$	$16 \times 16 \times 32$	2
Block 3	$16 \times 16 \times 32$	$8 \times 8 \times 64$	2
Block 4	$8 \times 8 \times 64$	$8 \times 8 \times 64$	1
Block 5	$8 \times 8 \times 64$	$8 \times 8 \times 64$	1
GAP	$8 \times 8 \times 64$	64	–

the multiply-accumulate operations of every layer in Table 3 gives ≈ 1.0 M^[VERIFY] MACs (about 2.0 MFLOPs) for one forward pass, of which the image branch accounts for 98%^[VERIFY]; the landmark and fusion paths together contribute only about 22 k MACs. The model is therefore cheap to run per frame in addition to being small to store, which matters for the edge setting where compute, not just disk size, is constrained.

4.3 Landmark branch and fusion

The landmark input preserves the order of the 21 MediaPipe points. Its crop-relative (x, y) coordinates are flattened in landmark order, producing 42 values. The landmark branch first projects $42 \rightarrow 128$ ^[ARCH], applies batch normalization and ReLU, then projects $128 \rightarrow 64$ ^[ARCH] followed by another normalization and ReLU. Unlike the image branch, this branch operates directly on finger and palm geometry and is unaffected by crop texture or color.

Let $\mathbf{f}_I \in \mathbb{R}^{64}$ and $\mathbf{f}_L \in \mathbb{R}^{64}$ denote the image and landmark descriptors. Fusion is performed by direct concatenation,

$$\mathbf{f} = [\mathbf{f}_I; \mathbf{f}_L] \in \mathbb{R}^{128}. \quad (3)$$

The fusion head applies a $128 \rightarrow 64$ ^[ARCH] linear layer and ReLU, then 0.3^[ARCH] dropout during training. A final $64 \rightarrow 6$ ^[ARCH] layer produces one logit for N/A and one for each target gesture. Softmax is applied only during evaluation and inference to convert these logits into class probabilities.

4.4 Conservative N/A rejection

The deployed decision procedure contains three safeguards^[INFER]. Before running the network, it computes the horizontal and vertical span of the 21 crop-relative landmarks. If the larger span is below 0.05^[INFER], all points have collapsed into a very small region and the sample is immediately rejected as N/A. For a valid sam-

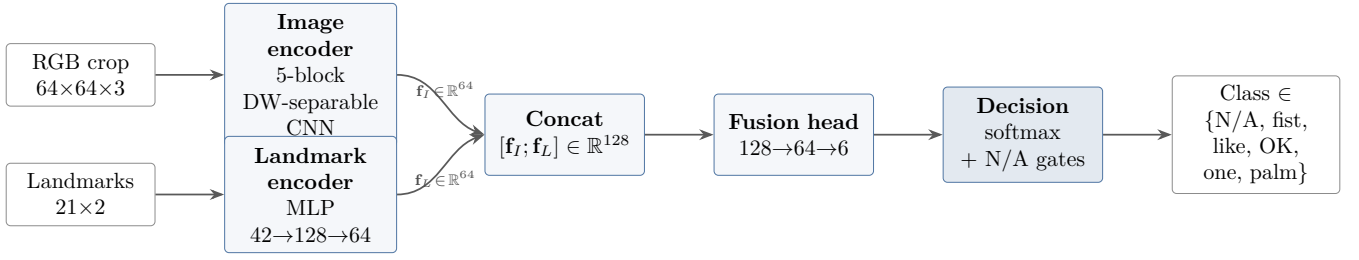


Figure 1: Overview of the dual-stream classifier. A MediaPipe-cropped RGB hand image and 21 crop-relative landmarks are encoded by two independent branches, each producing a 64-dimensional descriptor. The descriptors are concatenated into a 128-dimensional multimodal feature, mapped to six logits by the fusion head, and converted to a final label by the softmax and N/A-rejection gates.

ple, the image and landmarks pass through the two branches and softmax produces six probabilities. If the highest-probability class is already class 0, the model returns N/A directly. Otherwise, let p_1 be the highest non-N/A prediction and p_2 the runner-up probability. The sample is rejected when

$$p_1 < \tau_c \quad \text{or} \quad p_1 - p_2 < \tau_m. \quad (4)$$

Calibration selected $\tau_c = 0.95^{\text{[CAL]}}$ and $\tau_m = 0.00^{\text{[CAL]}}$. Therefore, the final deployed sequence is: landmark validity check, six-class neural prediction, direct neural N/A decision, and finally the 0.95 confidence gate, as shown in Figure 2. The margin condition remains implemented for reproducibility and future recalibration, but a zero threshold means it does not reject additional samples in the submitted configuration.

5 Dataset and Preprocessing

5.1 Dataset construction

We use the HaGRIDv2 512-pixel dataset [3]. The source dataset IDs 2, 9, 14, 15, and 16 are mapped respectively to fist, like, OK, one, and palm, which become output classes 1–5. Every retained one-handed category outside this set is assigned class 0 (N/A). Source IDs {5,6,7,8,23,30,33}^[DATA], which represent two-handed categories, are excluded rather than converted to N/A because the challenge classifier receives a single detected hand. This policy makes the N/A class a broad collection of non-command one-hand poses instead of a single semantic gesture. The exact mapping constants are retained in the submitted preprocessing source code.

5.2 MediaPipe crop generation

The preprocessor detects at most one^[DATA] hand using detection, presence, and tracking confidence thresholds of 0.5^[DATA]. Given the 21 image-relative landmarks, it finds their minimum and maximum image coordinates to form a tight bounding box. Padding equal to 30%^[DATA] of the larger box dimension is added on every side, and the padded box is clamped to the image boundary. The RGB pixels inside this box form the classifier crop.

Landmarks are then converted from full-image coordinates to crop-relative coordinates. For a crop (l, t, r, b)

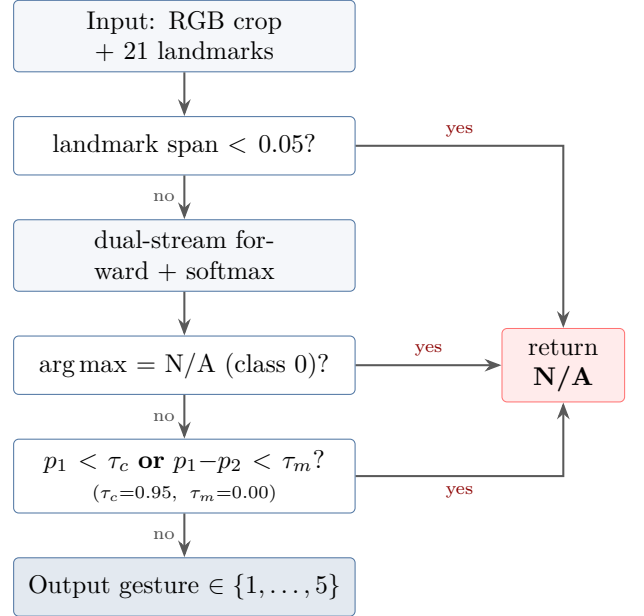


Figure 2: Deployed N/A-rejection decision flow^[INFER]. A sample is rejected as N/A if the landmarks collapse, if the network’s top class is N/A, or if it fails the confidence gate; otherwise the predicted gesture is returned. The margin term is inactive at $\tau_m=0.00$.

in an image of width W and height H , the transformation is

$$x_c = \frac{xW - l}{r - l}, \quad y_c = \frac{yH - t}{b - t}. \quad (5)$$

Images for which MediaPipe detects no hand are discarded. For each accepted sample, the pipeline stores the crop as JPEG and the 21×2 landmark array as a NumPy file; the original full-frame image is not retained. Figure 3 shows representative preprocessed samples with their landmarks overlaid, illustrating the two modalities the classifier receives.

Table 4 shows a substantial imbalance: the N/A split is approximately eight times the size of an individual target class^[PRES]. Phase 1 therefore uses inverse-frequency sampling, while Phase 2 uses inverse-frequency loss weights.

5.3 Landmark-synchronized augmentation

The official test applies reasonable perturbations such as bounding-box jitter and blur [1]. Training therefore applies $\pm 10\%$ ^[AUG] crop jitter, Gaussian blur with

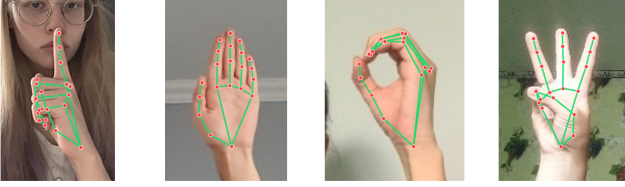


Figure 3: Representative preprocessed inputs. Each panel shows a MediaPipe-cropped RGB hand (the image-branch input) with its 21 crop-relative landmarks overlaid as the MediaPipe skeleton (the landmark-branch input). Examples are drawn from `sample_preprocessed_outputs/`^[DATA].

Table 4: Processed split distribution reported by the team presentation. Counts reflect samples retained after preprocessing.

Split	N/A	fist	like	OK	one	palm	Total
Train	487094	21079	20421	21646	21267	22196	593703
Val.	58407	2669	2625	2816	2661	2804	71982
Test	99385	4602	4568	4808	4633	4784	122780

Source: Team 14 presentation ^[PRES]. The original machine-readable count log was not committed.

$\sigma \in [0.1, 1.5]$ ^[AUG], brightness/contrast/saturation jitter of $\pm 20\%$ ^[AUG], horizontal flip with probability 0.5^[AUG], and rotation up to $\pm 15^\circ$ ^[AUG]. Spatial changes update the landmarks together with the image. For example, rotation around the crop center uses

$$x' = (x - .5) \cos \theta - (y - .5) \sin \theta + .5, \quad (6)$$

$$y' = (x - .5) \sin \theta + (y - .5) \cos \theta + .5. \quad (7)$$

This prevents image-landmark disagreement during multimodal training.

6 Training and Calibration

6.1 Phase 1: image pretraining

The first stage trains the image branch with a temporary $64 \rightarrow 6$ ^[P1] head for 20^[P1] epochs. We use Adam [8] with learning rate 10^{-3} ^[P1], batch size 256^[P1], cross-entropy loss, and cosine annealing. A `WeightedRandomSampler` assigns each class equal expected sampling weight. The project progress record reports a best validation accuracy of 86.15%^[PROG].

6.2 Phase 2: joint optimization

The best Phase-1 image weights initialize the full model. Joint training runs for 30^[P2] epochs with batch size 256^[P2]. The pretrained image branch uses learning rate 10^{-4} ^[P2], while the new landmark and fusion layers use 5×10^{-4} ^[P2]. The loss uses inverse-frequency class weights, with the N/A weight multiplied by 1.5^[P2] to encourage conservative decisions. Checkpoints are selected by validation contest score rather than accuracy alone. The reported best validation accuracy is 96.75%^[PROG].

6.3 Threshold calibration

After training, softmax outputs for the validation set are cached and scored over a $14 \times 11 = 154$ ^[CAL] grid. Confidence thresholds span 0.30–0.95^[CAL]; margin thresholds

span 0.00–0.50^[CAL]. Figure 4 shows that the best validation contest score, 11,936^[HEAT], occurs at the conservative confidence threshold of 0.95.

7 Results

7.1 Validation and model size

The architecture contains 36,039^[VERIFY] trainable parameters. There are two relevant size measurements. Summing model parameters and registered buffers gives 0.142 MiB^[VERIFY], the value shown in the architecture slides. The serialized `weights.pt` file is 176,372 bytes (0.168 MiB)^[VERIFY] because PyTorch serialization adds metadata. The latter is consistent with the leaderboard model-size score of 29.5/30^[PRES]. Distinguishing these quantities resolves the apparent difference between the presentation’s “0.142 MB” model statistic and the submitted file size.

7.2 Leaderboard performance

Run 1 achieved the stronger overall subtotal because its 1.5-point robustness advantage outweighed Run 2’s 0.6-point basic-performance advantage. Run 2 reached the maximum reported basic-performance score, but the lower robustness score suggests that model selection must account for real-world N/A examples rather than relying only on the HaGRIDv2-derived test set.

7.3 Baseline comparison: the N/A training distribution

To isolate how the N/A class should be constructed, we compare the submitted model against an earlier team baseline^[BASE] that shares the same dual-stream concept—a CNN image branch, a landmark MLP, and a fused classifier—but differs in two respects. First and most importantly, the baseline draws its N/A training data from only five specific HaGRIDv2 categories (`dislike`, `peace`, `stop`, `call`, and `no_gesture`), each capped at 4,000^[BASE] images, whereas the submitted model treats *every* retained non-command one-hand category as N/A (about 487k training samples; Section 5). Second, the baseline uses plain 3×3 convolutions on a 128×128 crop with a single 0.60^[BASE] confidence threshold for rejection, instead of the depthwise-separable blocks and multi-stage rejection of Section 4.

Table 7 reports both models under the official rule. The narrow-N/A baseline scores only 7.6/20 on basic performance and 5.0/40 on robustness, against 19.4 and 32.5 for the submitted model. The robustness gap is the most telling: a classifier exposed to only five N/A gesture types in training has no representation of the broad space of everyday hand poses in the TA-shot interference set, so it false-triggers frequently—and under the $+1/-2$ rule each false trigger is doubly costly. The two models also differ architecturally, so this is a baseline comparison rather than a single-variable ablation; nonetheless, the dominant cause of the gap is the N/A distribution. Defining N/A as the full diversity of

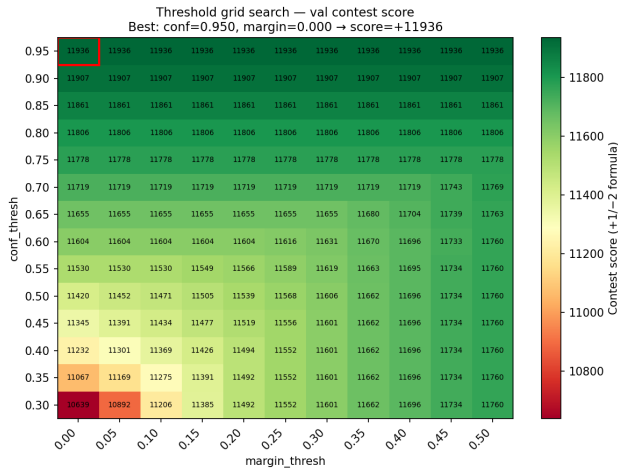


Figure 4: Validation contest score over the calibration grid. Each cell applies one confidence-margin pair to the same cached validation probabilities and evaluates the official $+1/-2$ objective.

Table 5: Training and calibration results reported by project artifacts.

Stage	Validation accuracy	Contest score
Image pretraining	86.15% ^[PROG]	—
Joint dual-stream	96.75% ^[PROG]	—
Calibrated model	—	11936 ^[HEAT]

non-command poses, rather than a handful of curated categories, is what makes the rejection behavior generalize to real-world inputs, and it is the main reason the submitted model is substantially more robust.

7.4 Modality ablation

To quantify each input stream’s contribution, we trained a landmark-only variant—the landmark branch followed by the same six-way head, with no image branch—under the same Phase-2 training protocol^[ABL]. Table 8 places it beside the image-only Phase-1 model and the full fusion model.

The landmark-only model reaches 96.60%^[ABL] validation accuracy with only 18,694 parameters—close to the full fusion model (96.75%^[PROG]) and far above the image-only model (86.15%^[PROG]). Hand geometry therefore carries most of the discriminative signal for these six classes on clean validation data. This ablation was run only after the fusion model had been finalized and submitted, on a freshly re-preprocessed copy of the dataset; the image-only and fusion figures are from the original build, so the small landmark-only vs. fusion accuracy gap should not be over-interpreted, although the large margin over image-only is robust to that difference.

We nonetheless keep the image branch in the submitted model, for robustness rather than clean-set accuracy. The landmark stream depends entirely on MediaPipe’s landmark estimates; when those degrade or fail—through occlusion, motion blur, unusual view-points, or detection errors that are common in real-world capture—a landmark-only classifier has no fallback, whereas the image branch supplies an independent appearance-based signal that can still yield a cor-

Table 6: Two official-evaluation runs reported in the presentation. “Total” is the subtotal before the 30-point presentation component.

Run	Size	Basic	Robustness	Total / 90
1	29.5	19.4	32.5	81.4
2	29.5	20.0	31.0	80.5

All values in this table are transcribed from the final presentation^[PRES].

Table 7: Effect of the N/A training distribution. Both models use the same dual-stream concept and the official $+1/-2$ scoring; they differ mainly in how the N/A class is built. Submitted scores are from the presentation^[PRES]; the baseline model and its scores are the team’s earlier run^[BASE].

Model	N/A training data	Basic /20	Robust. /40
Submitted	all non-command one-hand poses ($\approx 487k$)	19.4	32.5
Baseline	5 categories, $\leq 4k$ each ($\leq 20k$)	7.6	5.0

rect prediction. This is consistent with the evidence we do have: the full model’s best validation contest score ($+11,539$ ^[PROG], default gates) exceeds the landmark-only model’s ($+11,233$ ^[ABL]) by more than the small accuracy gap suggests, indicating the image branch already helps on borderline cases under the asymmetric scoring. A controlled real-world comparison of the two would quantify this robustness benefit directly and remains future work.

8 Discussion and Limitations

The results support three practical conclusions. First, a small custom network is sufficient for a high challenge score: the submission uses less than 0.2 MiB of serialized weights while retaining strong target-set performance. Second, calibration matters because the evaluation heavily penalizes false activation. The selected threshold of 0.95 deliberately trades recall for precision. Third, the baseline comparison in Section 7.3 shows that the N/A training distribution, not the network capacity, dominates real-world robustness: replacing a few curated N/A categories with the full diversity of non-command poses raises the robustness score from 5.0 to 32.5. The remaining gap between basic and robustness scores still indicates a residual domain shift between HaGRIDv2 imagery and TA-shot daily-action examples.

The available evidence also has limitations. The repository contains the training and calibration code, final weights, thresholds, and heatmap, but not the original Phase-1/Phase-2 logs or per-example leaderboard predictions. Consequently, the validation accuracies are traceable to the project progress record, while the leaderboard table is traceable to the team presentation; neither was independently recomputed in this report. The modality ablation (Section 7.4) shows that the landmark stream alone nearly matches the full model, but image-only and landmark-only were measured on slightly different data builds; a fully controlled single-build ablation of all three configurations remains future

Table 8: Modality ablation (best validation accuracy). Image-only and fusion are the original Phase-1/Phase-2 runs^[PROG]; landmark-only is a dedicated run on a freshly re-preprocessed copy of the dataset under the same protocol^[ABL]. The two data builds are not identical, so the landmark-only vs. fusion difference should be read as approximate.

Configuration	Params	Val. accuracy
Image-only (Phase 1)	13,639	86.15%
Landmark-only	18,694	96.60%
Fusion (image + landmark)	36,039	96.75%

work. Finally, latency and energy consumption were not measured on a specified edge device; small model size alone does not establish real-time performance on every deployment target.

9 Reproducibility and Evidence

Table 9 maps every evidence marker used beside a quantitative claim to source code, a generated artifact, the official specification, or the final presentation. The submitted report folder includes the presentation PDF and a verification script for model statistics.

10 Conclusion

We developed a compact dual-stream hand-gesture classifier that combines RGB appearance with MediaPipe landmark geometry and uses calibrated rejection to address asymmetric false-trigger costs. The final submission contains only 36,039 trainable parameters, and its strongest reported leaderboard run scored 81.4 out of 90 before presentation points. The remaining performance gap is primarily in real-world robustness, motivating future work on hard-negative collection, explicit modality ablation, and evaluation on measured edge hardware.

Code Availability

The complete inference and training source code, final model weights, threshold configuration, and environment requirements are available at https://github.com/xxjustintwxx/Multimedia_Final_Project. The report package includes the exact presentation, specification, figures, and verification helper used to support its claims.

Table 9: Evidence ledger for implementation details and quantitative claims.

Key	Authoritative source	Claims supported
SPEC	source/Final_Hand_Gesture_specs.pdf; Ref. [1]	Inputs, class definitions, limits, scoring rules, and test composition.
INFER	inference.py	Input normalization, label mapping, thresholds, and rejection logic.
ARCH	model/architecture.py	Layer dimensions, feature dimensions, and dropout.
AUG	train/augment.py ; train/dataset.py	Augmentation types, probabilities, and parameter ranges.
P1	train/train_phase1.py	Phase-1 architecture and hyperparameters.
P2	train/train_phase2.py	Phase-2 optimization, loss weighting, and checkpoint selection.
CAL	train/calibrate.py ; model/thresholds.json	Grid definition and deployed thresholds.
HEAT	figures/threshold_heatmap.png ; train/plot_threshold_heatmap.py	Grid-search scores and selected calibration point.
VERIFY	evidence/verify_report_numbers.py ; evidence/compute_flops.py ; model/architecture.py ; model/weights.pt download_and_preprocess.py ;	Parameter count, per-branch parameters, tensor footprint, checkpoint size, computed size score, and the MAC/FLOP inference cost.
DATA	hand_preprocess.py ; figures/make_sample_inputs.py ; sample_preprocessed_outputs/	Dataset source, label mapping, exclusion policy, preprocessing, and the rendered sample-input figure.
BASE	evidence/baseline/ ; download_dataset.sh , subsample_dataset.py , baseline_inference.py train/ablation_landmark_only.py	Baseline N/A category selection, the 4,000-per-class cap, baseline architecture and 0.60 threshold, and its leaderboard scores (team-reported; raw export not committed).
ABL		Landmark-only ablation: model definition, parameter count, and its best validation accuracy and contest score (run on a re-preprocessed copy of the dataset under the Phase-2 protocol).
PROG	PROGRESS.md	Recorded Phase-1 and Phase-2 validation accuracies and the selected training run. Original training logs are not present in the repository.
PRES	source/Hand Gesture Classification on Edge Devices.pdf; Ref. [9]	Retained split counts and leaderboard results. The original evaluation export or screenshot is not present in the repository.

References

- [1] Multimedia Course Teaching Team and Microsoft, “Hand gesture classification on edge devices: Challenge specification,” 2026, course project specification, local file: Final_Hand_Gesture_specs.pdf.
- [2] F. Zhang, V. Bazarevsky, A. Vakunov, A. Tkachenka, G. Sung, C.-L. Chang, and M. Grundmann, “Mediapipe hands: On-device real-time hand tracking,” *arXiv preprint arXiv:2006.10214*, 2020. [Online]. Available: <https://arxiv.org/abs/2006.10214>
- [3] A. Nuzhdin, A. Nagaev, A. Sautin, A. Kapitanov, and K. Kvanchiani, “HaGRIDv2: 1m images for static and dynamic hand gesture recognition,” *arXiv preprint arXiv:2412.01508*, 2024. [Online]. Available: <https://arxiv.org/abs/2412.01508>
- [4] A. G. Howard, M. Zhu, B. Chen, D. Kalenichenko, W. Wang, T. Weyand, M. Andreetto, and H. Adam, “MobileNets: Efficient convolutional neural networks for mobile vision applications,” *arXiv preprint arXiv:1704.04861*, 2017. [Online]. Available: <https://arxiv.org/abs/1704.04861>
- [5] S. Ioffe and C. Szegedy, “Batch normalization: Accelerating deep network training by reducing internal covariate shift,” in *Proceedings of the International Conference on Machine Learning*, 2015, pp. 448–456. [Online]. Available: <https://proceedings.mlr.press/v37/ioffe15.html>
- [6] M. Sandler, A. Howard, M. Zhu, A. Zhmoginov, and L.-C. Chen, “MobileNetV2: Inverted residuals and linear bottlenecks,” in *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition (CVPR)*, 2018, pp. 4510–4520. [Online]. Available: <https://arxiv.org/abs/1801.04381>
- [7] A. Howard, M. Sandler, G. Chu, L.-C. Chen, B. Chen, M. Tan, W. Wang, Y. Zhu, R. Pang, V. Vasudevan, Q. V. Le, and H. Adam, “Searching for MobileNetV3,” in *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, 2019, pp. 1314–1324. [Online]. Available: <https://arxiv.org/abs/1905.02244>
- [8] D. P. Kingma and J. Ba, “Adam: A method for stochastic optimization,” in *International Conference on Learning Representations*, 2015. [Online]. Available: <https://arxiv.org/abs/1412.6980>
- [9] Team 14, “Hand gesture classification on edge devices,” Jun. 2026, final project presentation, June 11, 2026.

Team Member Contributions

Table 10: Team member contribution statement.

Member	Student ID	Specific contributions
王昱閔	112062114	Model architecture design; model training and hyperparameter tuning; presentation; report writing.
王辰竹	112062105	Hyperparameter tuning; presentation.
羅詠璿	112060002	Baseline model training; real-time demo; report writing.